Yalamanchili Harsha Vardhan
Computer Science & Engineering
Indian Institute of Technology Bombay

24B1069
B.Tech.
Gender: Male
DOB: 03/01/2007

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	9 **Core:** 62 | **Total:** 62
Intermediate	CBSE	Narayana English Medium School	2024	97.80%
Matriculation	CBSE	Narayana High School	2022	97.60%

Scholastic Achievements

- Attained **All India Rank 170** in the **Joint Entrance Examination Advanced**, among **180,000+** aspirants *2024*
- Secured **Rank 84** in **TS EAPCET**, a premier state-level entrance exam, among **254,000+** contenders *2024*
- Achieved **All India Rank 267** in the **Joint Entrance Examination Main**, among **1,400,000+** aspirants *2024*
- Secured **Rank 119** in **AP EAPCET**, a major state-level entrance test, out of **339,000+** participants *2024*

Key Projects

Summer Of Quant | *Quant Community, IIT Bombay* *(June - July 2025)*

- Studied **Markov chains**, **stochastic** and **Poisson processes** with applications in quantitative modeling
- Gained exposure to **time series analysis**, **feature engineering**, and signal generation for trading strategies
- Learned **quantitative trading strategy design**, focusing on risk management, **backtesting**, and model robustness
- Secured a **Certificate of Excellence** and ranked among the top 10 performers in the Summer of Quant program

Cricket Scoring System | *Course Project* *(March - April 2025)*

Guide: Prof. Kameswari Chebrolu - Software Systems Lab

- Developed a real-time **Cricket Scoring Web Application** using **HTML, CSS, JavaScript, and Python**
- Implemented modular features for player management, score tracking, strike rotation, and over summaries with robust error handling, and used **Selenium** to automate webpage simulation for testing and input automation
- Integrated interactive **modals** for player input, live updates, and user-friendly match simulation experience

AI Chess Engine | *Self Project* *(June - July 2025)*

- Understood several core concepts of **Game Theory** such as **MinMax Theorem**, **Nash Equilibrium** and **Zero Sum Games** to develop optimal strategies for a wide variety of competitive scenarios and decision-making problems
- Designed an **Advanced Evaluation Function** for **Chess Analysis** incorporating current positional insight and future position forecast to make a **chess bot** capable of solving various **puzzles** with **90%+ accuracy** efficiently
- Extended chess bot capabilities to play complete matches, integrating features like **Opening Book Lookup**, **MinMax Theorem**, and **Alpha-Beta Pruning Optimization**, successfully defeating a **1500-rated** Lichess bot

Stock Price Prediction Using LSTM | *Self Project* *(Dec 2024)*

- Built a stock price prediction model using **LSTM networks** to capture temporal dependencies in past market data
- Collected, cleaned, and visualized stock data with **Pandas, NumPy, Matplotlib, yfinance**, and **Scikit-learn**
- Implemented real-time stock data collection and performed advanced **feature engineering** with **yfinance**

Advanced Algorithm Implementation | *Course Project* *(Aug - Sept 2025)*

Guide: Prof. Ashutosh Gupta - Data Structures and Algorithms

- Implemented a **Red-Black Tree** with insert/delete operations and an optimized **hash table** using linear probing
- Devised **Strassen's matrix multiplication** with advanced memory optimizations, improving its efficiency
- Investigated and benchmarked **Heaps**, **Binary Trees**, and **BSTs** to evaluate efficiency and operational trade-offs

Restaurant E-Commerce Website | *WebDev Project* *(Jan - Feb 2025)*

Devcom, IIT Bombay

- Built a full-stack **restaurant web app** with an interactive digital menu and real-time dynamic cart system
- Developed a fully responsive frontend in **React JS** and backend in **Node.js**, gaining hands-on experience in **modular web development**, efficient state management, and scalable e-commerce application design

Applied Crypto & Exploits | *Course Project* *(Aug - Sept 2025)*

Guide: Prof. Shruthi Shekar - Introduction to Cryptography

- Exploited a **CTR-mode** keystream reuse flaw in an echo-server using **PyCryptodome** to recover the secret flag
- Performed a **timing side-channel** attack on **HMAC verification**: exploited non-constant-time comparisons to recover MAC bytes by amplifying micro-timing differences through sampling; recommended constant-time functions
- Implemented the **Nostradamus diamond-structured** collision for Merkle–Damgård hashes: precomputed a parallelisable diamond to enable CTFP messages and demonstrate trade-offs in precomputation and storage

Other Projects

Competitive Programming | *Summer Of Science* *(June - July 2025)*
Maths and Physics Club, IIT Bombay

- Explored recursion, **dynamic programming**, string operations, **greedy methods**, and **mathematical algorithms**
- Solved over **150+** Codeforces problems and puzzles across diverse difficulty levels to build strong analytical thinking
- Applied advanced data structures in coding challenges, emphasizing efficiency and modularity in implementation
- Worked with graph algorithms (**Dijkstra's, Floyd-Warshall**) and **sorting techniques** to optimize performance

Quadcopter Design and Control | *Course Project* *(March - April 2025)*
Guide: Prof. Joseph John, Prof. Shankar Krishnan - Makerspace

- Developed a fully operational **quadcopter** from the ground up, combining structural design, electronics, and embedded coding for autonomous flight, while modeling essential parts in **Fusion 360** for accurate assembly
- Designed a **flight controller** with wireless connectivity to ensure stable control of orientation and movement
- Gained hands-on expertise in **aerodynamics, sensors, circuit design, and microcontroller programming**

Verilog and x86 Assembly | *Course Project* *(Aug - Oct 2025)*
Guide: Prof. Biswabandan Panda - Digital Logic and Computer Architecture Lab

- Developed a hardware-level **AES encryption** module in Verilog with optimized **data paths** and **control logic**
- Implemented supporting **sequential** and **combinational circuits** and mastered **x86 assembly**, efficiently managing **registers**, **memory**, and program flow, while engineering **structs** for precise and efficient low-level data layouts
- Applied **low-level programming** skills to implement mathematical algorithms in x86 assembly, including **complex number arithmetic, polar coordinate conversions**, and a **Sieve of Eratosthenes** prime generation algorithm

SAT-Based Sokoban Solver | *Course Project* *(Aug 2025)*
Instructor: Prof. S. Krishna - Logic in Computer Science

- Designed a **robust multi-dimensional state-space encoding** using **PySAT**, translating Sokoban game mechanics into **Boolean satisfiability** with 6+ constraint types, including movement logic and non-overlap conditions
- Enhanced solver performance using **unit propagation** and **clause learning**, while optimizing variable encoding to efficiently handle complex **SAT** instances and enable faster solution discovery and more scalable solutions
- Constructed **CNF** clauses modeling **collision dynamics** and **transition logic**, enabling valid solution paths

Applied AI Learning Project | *Alcheringa, IIT Guwahati* *(Jan - Feb 2025)*

- Completed an **8-week AI program**, gaining hands-on experience with modern AI and machine learning techniques
- Gained strong expertise in **machine learning algorithms, data processing, and predictive modeling**, and applied these advanced technical skills effectively to analyze data, build models, and solve real-world problems
- Demonstrated a strong and consistent commitment to continuous learning and advanced technical skill development

Innovative Python Tools & Applications | *Self Projects* *(June 2024 - Present)*

- Engineered an automated **Sudoku solver** with **OpenCV**, using **image processing** and **backtracking** to handle noisy input efficiently, accurately interpreting grids and solving challenging Sudoku puzzles in real-time applications
- Developed a Windows utility to adjust the opacity of any running application for improved multitasking and focus
- Built a Python script to fetch and display all problems solved by a user on **Codeforces**, tracking progress efficiently
- Created a PDF converter that transforms documents into a dark-themed format, reducing eye strain for extended use

Courses Undertaken

Computer Science	Discrete Structures, †Software Systems Lab, Computer Programming and Utilization, †Data Structures & Algorithms*, †Digital Logic Design and Computer Architecture*, Logic for Computer Science*, Data Analysis & Interpretation*, Introduction to Cryptography*
Mathematics	Calculus, Linear Algebra & Differential Equations
Others	Entrepreneurship, Philosophy, Makerspace, Biology, Chemistry Lab, Physics Lab, Economics*

†*Course has corresponding lab* **To be completed by Dec 2025*

Technical Skills

Languages	C/C++, Python, Bash, Assembly, Verilog, HTML, CSS, JavaScript, SQL, R, Java
Software & Tools	Git, GitHub, Docker, Make, LaTeX, Markdown, Sed, Awk, Jupyter, ReactJS, Fusion360
Crypto/Security	PyCryptodome, Hash Functions, Side-Channel Analysis, Constant-Time Programming
Data/ML Libraries	NumPy, Pandas, MatPlotLib, SciPy, TensorFlow, PyTorch, OpenCV, Scikit-learn
Embedded Systems	Arduino, FPGA (Verilog), Circuit Design, PCB Design, IMU/Gyroscope Sensors

Extracurriculars

- Volunteered with the **NSS**, actively supporting environmental initiatives and local community development projects
- Assisted in organising the **Treasure Hunt** during **E-Summit 2025**, managing logistics and engaging participants
- Acted as BlinkIt founders in an **ENT101** startup project, creating a pitch deck and presenting strategy to investors
- Represented school in **Volleyball**, developing teamwork, resilience, and discipline through tournament participation
- Participated in **EnB Buzz '24**, a competitive Business Model competition organized by **E-Cell, IIT Bombay**.